

G06 / ASSESSMENT

G06 GATE A

Standalone Zero-to-Job-Ready Data Engineer 2026

Study, practise, check and explain.

A complete learning route

Start with the lesson video. Read the worked explanation beside it, pause to reproduce the example, then solve the supplied task. The books contain your study notes; keep your own work and results as evidence.

Use the protected answers after saving your first attempt. A changed retry shows what you can now do independently. There is no forced timer during learning.

One course. M00-M19. 264 lessons and 50 project phases.

Contents

Tap a topic or use the PDF bookmarks. These numbers are the actual pages in this edition.

Read this first 3

Case 3

Read this first

Protected stage-boundary assessment for M15-M17: Cloud + Production Engineering + Architecture.

Rule: Treat this as unseen. Save the original answer and evidence before opening the matching protected review. Gate B is a different fresh case, not a repeat of Gate A.

Case

You are reviewing a cloud data platform for a regional insurer. Claims arrive as hourly files (25 GB/day). Fraud rules can wait 15 minutes, executive reporting can wait 2 hours. Accepted raw evidence must never be lost. PII/health data requires restricted access. The data team has five engineers and no dedicated Kubernetes platform team. Current cost target is 18,000 units/month. The old design copies full raw data into three systems and uses one shared admin service account.

1. Extract functional/non-functional requirements, unknowns and constraints. Label assumptions.
2. Draw a data architecture and state batch/stream, warehouse/lakehouse and managed/build choices.
3. Define RTO/RPO, failure domains, replay/checkpoint/publish behavior and one deliberate failure test.
4. Create least-privilege identity/data-access matrix and sensitive-data boundary.
5. Identify the bottleneck under current volume and under 2x volume; show one capacity calculation.
6. Estimate major monthly cost drivers and one safe optimization; do not remove raw/recovery controls.
7. Add CI/CD/rollback/observability controls from M16 and explain evidence required before promotion.
8. Promotion map: say whether CDC, data contracts, Vault, data mesh or Kubernetes are justified here and why.
9. Runnable integration evidence: execute at least two supplied local M15-M17 simulators/scorecards and attach output. Deliberately fail one control, repair it, then run a changed-input retry.
10. Defend the architecture in five minutes: requirement -> decision -> evidence -> trade-off -> failure behavior.

Critical failures: no saved first attempt; no explicit recovery/replay; broad admin identity; architecture choice with no requirement; claiming live cloud/runtime evidence that was not executed.